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MODULE - 4

* PARTIAL ORDERED RELATION

$aRa \Rightarrow$ Reflexive

$aRb, bRa \Rightarrow$ Symmetry

$aRb, bRa \Rightarrow a=b \Rightarrow$ Anti-Symmetry

$aRb, bRc \Rightarrow aRc \Rightarrow$ Transitive

A relation is partial ordered if it satisfies, reflexive, anti-symmetry and transitive properties. *

NOTE:

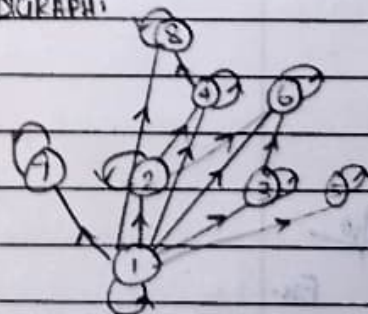
Then set A wrt partial ordered relation R is called POSET.

Ex:

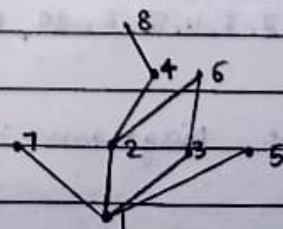
Draw the digraph representing the partial ordering, $\{(a,b) \mid a \text{ divides } b\}$ on the set $\{1, 2, \dots, 8\}$. Reduce it to Hasse Diagram.

$R = \{ (1,1), (1,2), \dots, (1,8) \\ (2,2), (2,4), (2,6), (2,8) \\ (3,3), (3,6), \\ (4,4), (4,8), \\ (5,5), \\ (6,6), \\ (7,7), \\ (8,8) \}$

DIGRAPH:



HASSE DIAGRAM:



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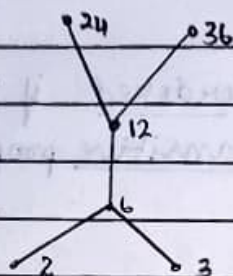
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2.

$$A = \{2, 3, 6, 12, 24, 36\}$$

$$R = \{(a, b) \mid a \text{ div } b\}$$

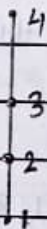
Hasse Diag:



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Draw Hasse diag for $(P, \geq)$

$$P = \{1, 2, 3, 4\} \quad \text{and} \quad \leq \text{ be 'less than eq to'}$$



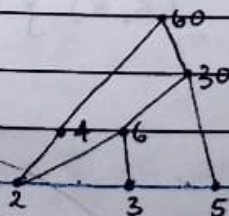
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Ex.

1.

$$(A = \{2, 3, 4, 5, 6, 30, 60\}, \text{div})$$

Hasse Diagram :



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2.

$$A = \{a\}$$

$$A = \{a, b\}$$

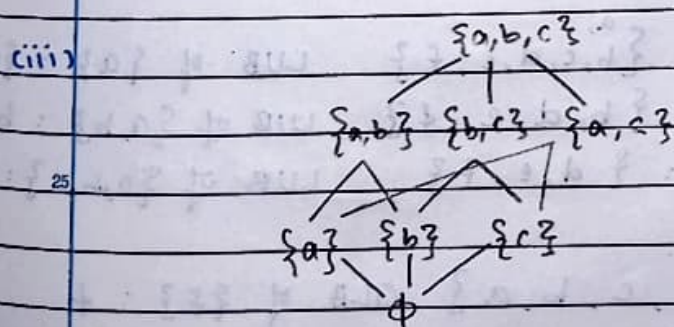
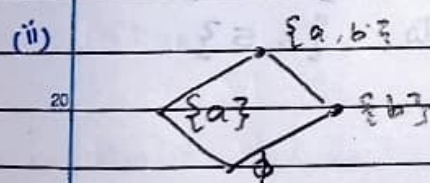
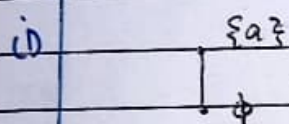
$$A = \{a, b, c\}$$

Find $P(A)$ and draw Hasse diagram of $(P(A), \subseteq)$
 $\hookrightarrow$ Power set of A . $\hookrightarrow$ containing

(i) $P(A) = \{\phi, \{a\}\}$

(ii) $P(A) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$

(iii) $P(A) = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$



* GLB, LUB

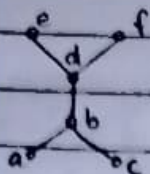
GLB : greatest lower bound (infimum)

LUB : least upper bound (supremum)

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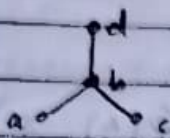
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eg:



min elements = $\{a, c\}$

max elements = $\{e, f\}$



min elements = $\{a, c\}$

max elements = $\{d\}$

$\therefore$ greatest element : d

least element : -

Ex:

1.

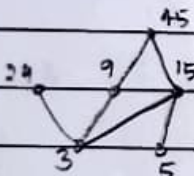
$A = \{3, 5, 9, 15, 24, 45\}$ with divisibility relation.

greatest element : -

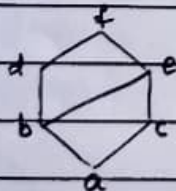
least element : -

max elements : $\{24, 45\}$

min elements : $\{3, 5\}$



eg:



UB of $\{a\}$: $\{b, c, d, e, f\}$

LUB of $\{a\}$: a

UB of $\{a, b\}$: $\{b, d, e, f\}$

LUB of $\{a, b\}$: b

UB of $\{a, b, c\}$: $\{e, f\}$

LUB of $\{a, b, c\}$: e

LB of $\{f\}$: $\{f, d, e, c, b, a\}$

GLB of $\{f\}$: f

LB of $\{f, d\}$: $\{d, b, a\}$

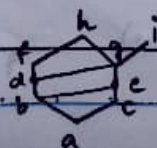
GLB of $\{f, d\}$: d

LB of $\{f, d, e\}$: $\{b, a\}$

GLB of $\{f, d, e\}$: b

Ex:

1.



GLB $\{b, d, f\}$: b

LUB $\{b, d, f\}$: f

LB $\{b, d, f\}$: $\{b, a\}$

UB $\{b, d, f\}$: $\{f, h, i\}$

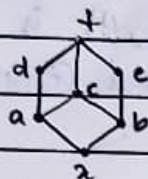
* LATTICE

A lattice is a POSET $(L, \leq)$ which has every two elements $a, b \in L$ has GLB and LUB.

GLB: $*$, $\wedge$, $\cdot$: product / meet

LUB: $\oplus$, $\vee$, $+$: sum / join

eg



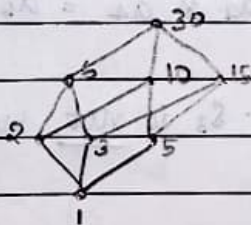
This is a lattice.

Ex

1.

$\langle D_{30}, 1 \rangle$ is lattice?

$$D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$$



greatest element : 30

lowest element : 1

$$2 * 5 = 1 ; 2 + 5 = 10$$

It is noted that for every $a, b \in D_{30}$ it satisfies $a * b$ and $a + b$. Hence it is a lattice.

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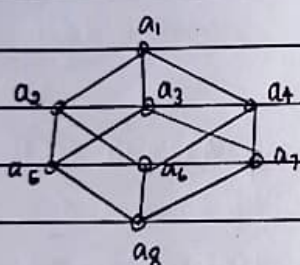
* PROPERTIES OF LATTICES

$$\begin{array}{llll}
 a * a = a & a \oplus a = a & \text{idempotent} & \\
 a * b = b * a & a \oplus b = b \oplus a & \text{commutative} & \\
 a * (b * c) = a * (b * c) & a \oplus (b \oplus c) = a \oplus (b \oplus c) & \text{associative} & \\
 a * (a \oplus b) = a & a \oplus (a * b) = a & \text{absorption} &
 \end{array}
 \left. \vphantom{\begin{array}{l} a * a = a \\ a * b = b * a \\ a * (b * c) = a * (b * c) \end{array}} \right\} \text{LATTICE}$$

* SUB-LATTICE

Let $\langle L, \oplus, * \rangle$ be a lattice. Let $S \subseteq L$, it is known as a sub-lattice iff it is closed on $*$ and $\oplus$.

Ex.



$$S_1 = \{a_1, a_4, a_2, a_6\}$$

$$S_2 = \{a_3, a_5, a_1, a_8\}$$

$$S_3 = \{a_1, a_2, a_4, a_8\}$$

S1:

$$a_1 + a_2 = a_1 \in S_1$$

$$a_1 * a_2 = a_2 \in S_1$$

S3:

$$a_2 * a_4 = a_6 \notin S_3$$

$\therefore S_3$ is not sub lattice.

$\therefore S_1$ is a sub-lattice.

S2:

$$a_3 * a_5 = a_5 \in S_2$$

$$a_3 + a_5 = a_3 \in S_2$$

$\therefore S_2$ is a sub-lattice.

If the POSET is infinite $\rightarrow$ it is a lattice but not bound

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* COMPLETE LATTICE

A complete lattice is such that every non-empty subset has GLB and LUB.

* BOUND LATTICE

A lattice which has both bound values $(0, 1)$ - (greatest / lowest element), then lattice is called bound lattice.

* COMPLEMENT ELEMENT

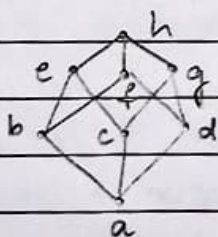
$$a * a' = 0$$

$$a + a' = 1$$

here a' is complement of a .

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If the complement of each element exists, then it is known as a complement lattice.



$$\text{here: } a' = h$$

$$b' = g$$

$$c' = f$$

$$d' = e$$

} all elements have complement

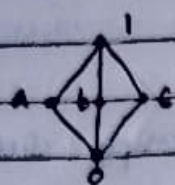
$\therefore$ complement lattice.

* DISTRIBUTIVE LATTICE

$$\left. \begin{aligned} a * (b + c) &= (a * b) + (a * c) \\ a + (b * c) &= (a + b) * (a + c) \end{aligned} \right\} \text{for any 3 elements}$$

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eg:



$$a * (b + c) = (a * b) + (a * c)$$

LHS:

$$a * (1) = a$$

RHS:

$$0 + 0 = 0$$

LHS $\neq$ RHS, thus it is not a distributive lattice.

* BOOLEAN ALGEBRA

A lattice which is complemented & distributive it is a boolean algebra.

* SUB BOOLEAN ALGEBRA

Let B be a boolean algebra and S be $S \subseteq B$. If S contains elements 0 and 1 and is closed under $\langle +, * \rangle$ then S is known as sub boolean algebra of B .

* PROPERTIES OF BOUNDS

$$a + 0 = a$$

$$a \cdot 1 = a$$

$$a + 1 = 1$$

$$a \cdot 0 = 0$$

* DE-MORGAN'S LAW

$$(a \cdot b)' = a' + b'$$

$$(a + b)' = a' \cdot b'$$

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NOTE:

$$a + a'b = a + b$$

proof :

$$\begin{aligned} & (a + a'b) \\ &= (a + a') \cdot (a + b) \\ &= a \cdot a + a \cdot b + a' \cdot a + a' \cdot b \\ &= a + a \cdot b + 0 + a'b \\ &= a + a'b \end{aligned}$$

$$\begin{aligned} & (a + a') \cdot (a + b) = a + a'b \\ & 1 \cdot (a + b) = a + a'b // \end{aligned}$$

Ex:

1.

$$\begin{aligned} & ((x_1 x_2)' x_3)' ((x_1' + x_2) (x_2' + x_3'))' \\ &= ((x_1 x_2) + x_3') \cdot ((x_1' + x_2)' + (x_2' + x_3'))' \\ &= (x_1 x_2 + x_3') \cdot (x_1 x_3' + x_2 x_3) \\ &= ((x_1 x_2 + x_3') \cdot (x_1 x_3')) + ((x_1 x_2 + x_3') \cdot (x_2 x_3)) \\ &= (x_1 x_2 x_3') + (x_1 x_3') + (x_1 x_2 x_3) + (x_2 x_3') \\ &= x_1 x_2 x_3' + x_1 x_2 x_3 + x_1 x_3' + x_2 x_3' \cdot x_3 \\ &= x_1 x_2 x_3' + x_1 x_2 x_3 + x_1 x_3' (x_2 + x_2') \\ &= x_1 x_2 x_3' + x_1 x_2 x_3 + x_1 x_2 x_3' + x_1 x_2' x_3' \\ &= x_1 x_2 x_3' + x_1 x_2 x_3 + x_1 x_2' x_3' // \end{aligned}$$

$$\therefore \Sigma m(6, 7, 4) //, \Sigma m(4, 6, 7) //$$

2.

$$\begin{aligned} & x_1 (x_2' x_3)' \\ &= x_1 (x_2 + x_3') = x_1 x_2 + x_1 x_3' \end{aligned}$$

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$$= x_1 x_2 x_3 + x_1 x_2 x_3' + x_1 x_2' x_3 + x_1 x_2' x_3'$$

$$= x_1 x_2 x_3 + x_1 x_2 x_3' + x_1 x_2' x_3' //$$

$$= \text{E.M.}(4, 6, 7) //$$

3.

$$x (y + z (xy + xz)')$$

$$= x \cdot (y + z \cdot (xy)' \cdot (xz)')$$

$$= x \cdot (y + z \cdot ((x' + y') \cdot (x' + z')))$$

$$= x \cdot (y + xz \cdot ((x' + y') \cdot (x' + z')))$$

$$= xy + (xy'z)(x' + z')$$

$$= xy + 0 = xy //$$

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4.

$$xy' + z + (x' + y)z'$$

$$= xy' + z + x'z' + yz'$$

$$= xy' + x' + z + yz'$$

$$= xy' + yz' + x' + z$$

$$= x' + y' + yz' + z$$

$$= x' + y' + y + z$$

$$= x' + z + 1 = 1 //$$

5.

$$x(y' + z')$$

$$= xy' + xz'$$

$$= xy'(z + z') + xz'(y + y')$$

$$= xy'z + xy'z' + xyz' + xy'z'$$

$$= xy'z + xy'z' + xyz'$$

$$= \text{E.M.}(4, 5, 6) //$$

$$x(y' + z')$$

$$= [x + (y \cdot \bar{y})] \cdot (y' + z')$$

$$= (x + y) \cdot (x + y') \cdot (y' + z')$$

$$= ((x + y) + (z\bar{z})) \cdot ((x + y') + (z\bar{z})) \cdot (y' + z') \cdot (x\bar{x})$$

$$= (x + y + z) \cdot (x + y + \bar{z}) \cdot (x + y' + z) \cdot (x + y' + \bar{z}) \cdot (x + y + \bar{z}) \cdot (x + y + z)$$

$$= (x' + y' + z') \cdot (x + y' + z') \cdot (x + y' + z) \cdot (x + y + z') \cdot (x + y + z)$$

$$= M_0 \cdot M_1 \cdot M_2 \cdot M_3 \cdot M_7$$

$$= \pi M(0, 1, 2, 3, 7) //$$

6.

$$a'b'c + ab'c + ab'c'$$

$$= b'c(a' + a) + ab'c'$$

$$= b'c + ab'c'$$

$$= (ac' + c)b'$$

$$= b'(a + c) = ab' + b'c //$$

7.20

$$a'b(a' + c) + ab'(b' + c)$$

$$= a'b + a'bc + ab' + ab'c$$

$$= a'b + ab' + c(a'b + ab')$$

$$= (c + 1)(a'b + ab') = a'b + ab' //$$

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8.

$$a + a'bc' + (b + c)'$$

$$= a + a'bc' + b' \cdot c'$$

$$= a + (a'b + b') \cdot c'$$

$$= a + \bar{c}(a' + b')$$

$$= a + \bar{a}\bar{c} + \bar{b}\bar{c}$$

$$= a + \bar{c} + b\bar{c} = a + \bar{c}(1 + b)$$

$$= a + \bar{c} //$$